From Bag-of-Words to Large Language Models: A Unified Comparative System for Text Classification

Jiahao Wu
Student ID: 2113454
College of Computer Science, Nankai University

Abstract

This proposal presents **ClassiScope**, a unified comparative system for text classification that integrates four model families—sparse lexical features, static word embeddings, fine-tuned pretrained transformers, and prompting of instruction-tuned large language models (LLMs)—under a single data protocol, evaluation suite, and web interface. The project is motivated by an observation from our course exercises: on news topic classification, a simple word-frequency baseline slightly outperformed a fine-tuned BERT under identical splits, raising the question of *when* model sophistication actually pays off. We plan to answer this question with controlled experiments on two public benchmarks, an interpretability module that compares token-level explanations across model families, and robustness and efficiency analyses. The deliverables are an open-source system with a live demonstration, a controlled comparison study, and an ACL-format final report.

1 Team Members and Responsibilities

This project is completed individually. Jiahao Wu (Student ID: 2113454) is solely responsible for all components: problem formulation and related-work survey, system design and implementation, data preparation, all experiments and analysis, the final report and presentation, and the runnable demonstration.

2 Background and Motivation

Text classification is the canonical task of NLP, and its solutions have evolved through four technical generations: (i) sparse lexical features with linear classifiers, such as bag-of-words or TF–IDF with logistic regression; (ii) distributed word embeddings (Mikolov et al., 2013; Pennington et al., 2014) compressed into document vectors and fed to lightweight classifiers; (iii) fine-tuned pretrained transformers (Vaswani et al., 2017; Devlin et al., 2019); and (iv) instruction-tuned LLMs applied through zero-/few-shot prompting, without any task-specific training (Brown et al., 2020; Ouyang et al., 2022; Qwen Team, 2024).

These families embody different trade-offs among labeled-data requirements, compute cost, interpretability, and deployment effort, yet published comparisons are often fragmentary: datasets, splits, tuning budgets, and metrics differ from study to study. Our own course exercise made the issue concrete. On the NYT news topic classification task, with identical 80/10/10 stratified splits for every model, a plain word-frequency baseline reached slightly *higher* accuracy (99.1%) than a fine-tuned BERT (98.4%). Sophistication, in other words, must be earned rather than assumed. This observation motivates the central question of the project—*under what conditions does each model family actually pay off?*—and the system contribution: a single, controlled, and demonstrable platform in which all families are trained, explained, stress-tested, and compared under exactly the same protocol.

3 Problem Definition

We will build **ClassiScope**, a unified text-classification system, and use it to answer four research questions:

- **RQ1 (Accuracy).** How do the four model families compare on the same datasets, with identical splits, preprocessing, and metrics?
- **RQ2 (Robustness).** How do their predictions degrade under character-level perturbations (typos, spacing, casing) and cross-task transfer?
- **RQ3 (Interpretability).** What token-level evidence does each family rely on, and to what extent do explanations agree across families?

- **RQ4 (Efficiency).** What accuracy–cost trade-offs (trainable parameters, training time, inference latency) arise for practical deployment?

4 Proposed System and Implementation Plan

System functions. Given an input text, **ClassiScope** produces side-by-side predictions from all four model routes with confidence scores, token-level explanation highlights, and model metadata; given a configured dataset, it trains or loads all routes, runs the robustness and efficiency suites, and renders a comparison dashboard. A web interface (Gradio) exposes the full pipeline as a live demonstration.

Architecture. The system has three layers. (1) A *data layer* with unified loaders and preprocessing; datasets are shuffled and split once (80/10/10, stratified, fixed seed) and all models consume the same cached splits. (2) A *model layer* in which four routes implement a common `predict_with_explanation` interface: TF–IDF + logistic regression; Word2Vec/GloVe mean-pooled document vectors + logistic regression; fine-tuned BERT-base; and an instruction-tuned LLM (Qwen2.5, accessed via API, with a small local variant as fallback) under zero-/few-shot prompting. (3) An *analysis layer* providing interpretability (logistic-regression coefficients; gradient-based saliency for BERT; LLM rationales), the robustness suite, an efficiency profiler, and the web demo.

Implementation plan. Existing course-exercise code for the first three routes will be ported into the framework and refactored behind the common interface; the LLM route, explanation module, and web demo are built new. All experiments are scripted, seeded, and reproducible; the source code will ship with an English README covering the environment, dependencies, data acquisition, and run steps.

5 Evaluation Plan

We evaluate on two public benchmarks spanning two classification tasks: the NYT news topic dataset used in course exercises (Sandhaus, 2008) and the SST-2 sentiment dataset (Socher et al., 2013). All routes report **Accuracy** and **Macro-F1** on the same held-out test set, with paired bootstrap tests for significance. Robustness is measured as accuracy deltas under the perturbation suite and as cross-task transfer gaps. Efficiency is reported as trainable parameters, wall-clock training time, and per-sample inference latency. Interpretability is assessed qualitatively and quantitatively, by the overlap of top-attributed tokens across families. The four routes serve as baselines for one another; the study itself is the comparison, so no external system is required for a valid result.

6 Expected Outcomes

The project will deliver: (1) an open-source, reproducible system with a runnable demo (live demo or video, per course requirements); (2) a controlled comparison study answering RQ1–RQ4; (3) an eight-page ACL-format final report; and (4) final slides and a contribution statement. We hypothesize that (H1) the fine-tuned transformer is strongest in-domain when labeled data is plentiful, with sparse and embedding-based baselines close behind on easy topic tasks, as our course results suggest; (H2) LLM prompting is competitive without any task-specific training, especially in low-data regimes, but at higher latency and cost; and (H3) the families make complementary errors, which their respective explanations expose. The system is designed so that whichever hypotheses the data overturns, the resulting analysis remains a substantive contribution.

7 Project Timeline

Phase / Period	Tasks
Phase 1: Proposal	This document (due Oct. 12, 2026).
Weeks 1–2 (Oct. 13–26)	Unified data pipeline; port sparse, embedding, and BERT routes onto both datasets; demo skeleton.
Phase 2: Midterm (date TBD)	All four routes running with preliminary results; midterm report and progress slides.
Weeks 3–5 (Oct. 27–Nov. 16)	LLM prompting route; interpretability, robustness, and efficiency studies; full analysis.
Weeks 6–7 (Nov. 17–30)	Final report writing; demo polish and demo video; final slides and contribution statement.
Phase 3: Final (dates TBD)	Final report, slides, source code, demo, and contribution statement submitted; in-class presentation.

References

Tom B. Brown et al. 2020. Language models are few-shot learners. In *Advances in Neural Information Processing Systems 33*.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*.

Tomas Mikolov, Kai Chen, Greg Corrado, and Jeffrey Dean. 2013. Efficient estimation of word representations in vector space. In *Proceedings of the First International Conference on Learning Representations (ICLR 2013), Workshop Track*.

Long Ouyang et al. 2022. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems 35*.

Jeffrey Pennington, Richard Socher, and Christopher D. Manning. 2014. GloVe: Global vectors for word representation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Qwen Team. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Evan Sandhaus. 2008. The new york times annotated corpus. LDC2008T19.

Richard Socher, Alex Perelygin, Jean Y. Wu, Jason Chuang, Christopher D. Manning, Andrew Y. Ng, and Christopher Potts. 2013. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. In *Advances in Neural Information Processing Systems 30*.